



Vietnamese-German University

Computer Network Group Project

Real-Time Light Intensity Monitoring System Using STM32 with Web-Based Interface

July 21, 2026

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Contents

1	Introduction	3
2	Sensor Description and Technical Background	3
2.1	BH1750 Digital Light Sensor	3
2.2	STM32 Microcontroller	4
3	System Architecture	5
4	User Interface	5
5	Software Implementation	6
5.1	STM32 Firmware	7
5.2	Python Serial Reader	7
5.3	Backend Server	8
5.4	Web-Based Frontend	9
6	Results and Discussion	10
6.1	Experimental Setup and Procedure	10
6.2	Real-Time Monitoring Results	12
7	Conclusion	13
8	ANNEX	14
8.1	read_sensor.py	14
8.2	server.js	15
8.3	home.js	17
8.4	home-WebSocket.js	17
8.5	home-DOM.js	19
8.6	home-plotly.js	19
8.7	home-time_zone.js	22
8.8	index.html	23
8.9	style.css	24
8.10	BH1750.h	30
8.11	BH1750.c	30
8.12	main.c	31

1 Introduction

The Internet of Things (IoT) has become an essential technology for monitoring and controlling physical environments through interconnected devices. In many IoT applications, sensor data must be acquired, transmitted, processed, and visualized in real time to provide users with immediate information about the monitored system. Developing a reliable end-to-end communication pipeline between embedded hardware and a graphical user interface is therefore an important aspect of modern embedded system design.

This project presents the implementation of a real-time light intensity monitoring system using a BH1750 digital light sensor and an STM32 microcontroller. The BH1750 sensor continuously measures ambient illumination, while the STM32 acquires the sensor data through the Inter-Integrated Circuit (I²C) communication protocol. The measured values are transmitted to a personal computer through a Universal Serial Bus (USB) Virtual COM Port.

On the software side, a Python application reads the incoming serial data, validates the received measurements, and converts them into JavaScript Object Notation (JSON) format. The JSON messages are then transmitted to a Node.js backend through a WebSocket connection. The backend server broadcasts the sensor data to one or more connected web clients, allowing multiple users to observe the sensor measurements simultaneously. Finally, a browser-based graphical user interface (GUI) displays the received illumination values and visualizes the data in real time using an interactive graph.

The objective of this project is to design and implement a complete data acquisition and visualization pipeline, beginning with physical sensing and ending with real-time presentation on a web interface. The system demonstrates the integration of embedded programming, serial communication, WebSocket networking, and web technologies to provide an efficient and scalable IoT monitoring solution.

2 Sensor Description and Technical Background

This section introduces the hardware components used in the proposed monitoring system. The BH1750 digital light sensor is responsible for measuring ambient illumination, while the STM32 microcontroller acquires the sensor data through the I²C communication interface and forwards the measurements to the host computer via USB Virtual COM Port. The technical characteristics and operating principles of both components are described in the following subsections.

2.1 BH1750 Digital Light Sensor

The BH1750 is a digital ambient light sensor developed by ROHM Semiconductor for measuring visible light intensity. Unlike analog photoresistors, the BH1750 integrates a photodiode, signal conditioning circuitry, a 16-bit analog-to-digital converter (ADC), and an I²C communication interface into a single integrated circuit. As a result, the sensor directly outputs calibrated digital measurements, eliminating the need for external analog-to-digital conversion.

The sensor measures illumination in units of lux (lx), where one lux corresponds to one lumen per square meter. The measured digital value is converted into illumination using the relationship provided in the manufacturer datasheet [1]

$$\text{Lux} = \frac{\text{Raw Output}}{1.2}. \quad (1)$$

In this project, the BH1750 operates in Continuous High Resolution Mode, allowing continuous measurements with a resolution of approximately 1 lux while automatically updating the measurement register. The STM32 periodically reads the two-byte measurement result through the I²C interface every second.

The BH1750 offers several advantages for embedded monitoring applications, including low power consumption, factory calibration, high measurement accuracy, and a simple two-wire communication interface. These characteristics make it well suited for real-time environmental monitoring systems.

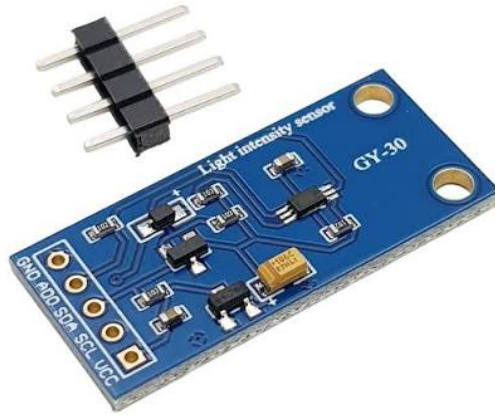


Figure 1: BH1750 digital ambient light sensor module.

2.2 STM32 Microcontroller

The embedded controller used in this project is an STM32 microcontroller from STMicroelectronics [2]. The microcontroller is responsible for configuring the BH1750 sensor, acquiring illumination measurements through the I²C peripheral, and transmitting the measured data to a personal computer through a USB Virtual COM Port (USB CDC).

The firmware was developed using STM32CubeIDE together with the STM32 Hardware Abstraction Layer (HAL). After system initialization, the firmware configures the BH1750 for continuous high-resolution measurements. During each sampling cycle, two bytes are read from the sensor and combined into a 16-bit raw measurement. The raw value is subsequently converted into lux according to the BH1750 conversion formula before being transmitted through the USB interface.

Using the USB CDC interface allows the STM32 board to appear as a standard serial (COM) port on the host computer, enabling reliable communication with the Python-based serial reader without requiring additional hardware such as a USB-to-UART converter.

3 System Architecture

The developed monitoring system consists of both hardware and software components that cooperate to acquire, transmit, process, and visualize ambient light measurements in real time. The overall architecture is divided into three layers: the sensing layer, the communication layer, and the application layer. The sensing layer is responsible for measuring the physical quantity, the communication layer transfers the measurements between different software modules, and the application layer presents the received data to end users through a web-based graphical user interface.

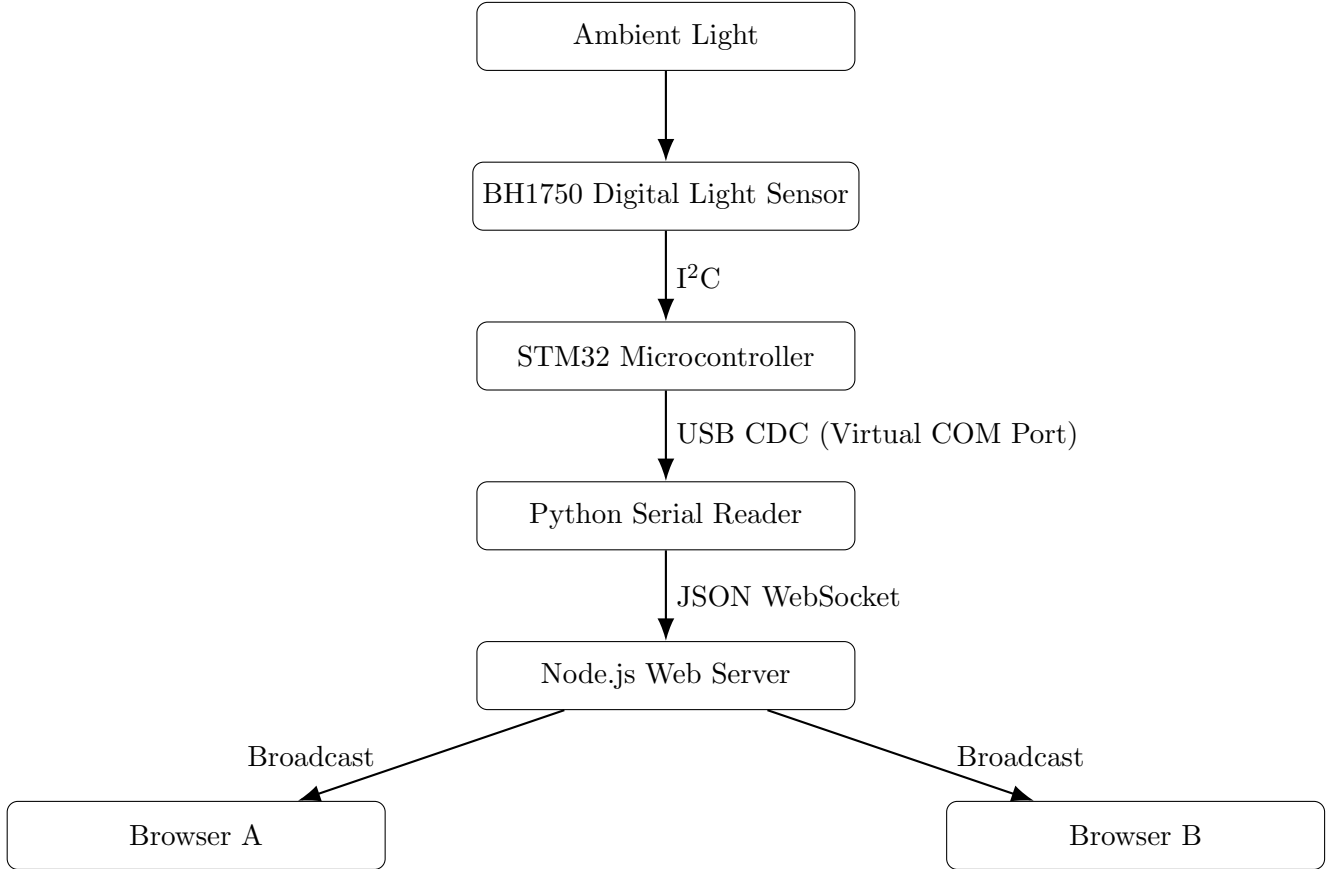


Figure 2: Overall system architecture of the proposed real-time light monitoring system.

4 User Interface

The proposed monitoring system provides a web-based graphical user interface (GUI) that allows users to observe ambient light measurements in real time. Unlike conventional desktop applications, a browser-based interface requires no software installation and can be accessed from any device connected to the network through a standard web browser. The interface receives sensor measurements from the Node.js backend using WebSocket communication, enabling continuous updates without requiring the user to manually refresh the page.

The graphical user interface is implemented using standard web technologies, including HTML, Cascading Style Sheets (CSS), and JavaScript.

Group 4: Light Intensity Monitoring

The interface contains three primary control buttons:

- **Run Sensor Data:** Establishes a WebSocket connection with the server and begins receiving real-time sensor data.
- **Stop:** Terminates the current streaming session and closes the WebSocket connection.
- **Download Graph:** Exports the current graph as an image file for documentation or further analysis, including PNG, SVG, and JSON.

In addition to these controls, the interface displays the current connection status, enabling users to determine whether the system is actively receiving sensor data or disconnected from the server.

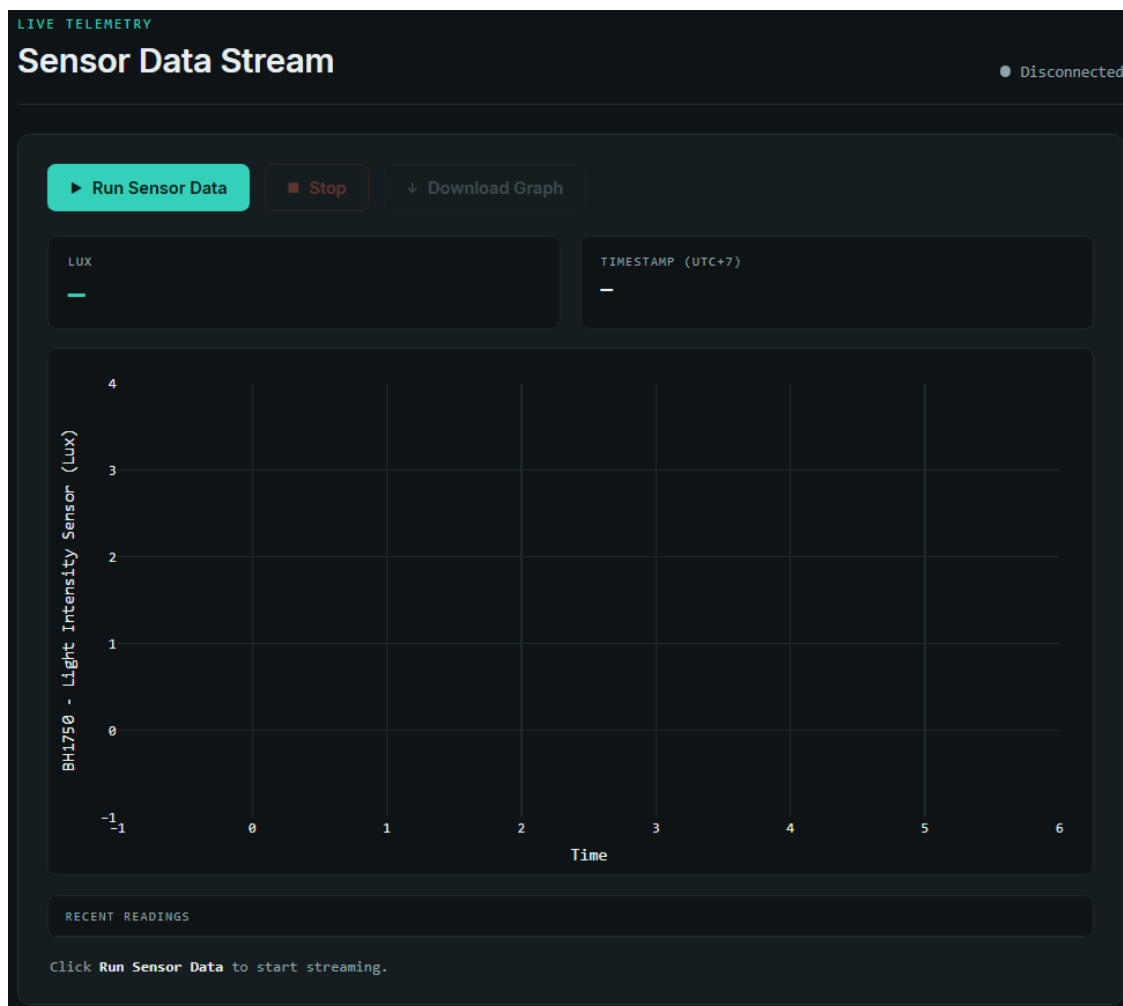


Figure 3: Web-based graphical user interface developed for real-time light monitoring.

5 Software Implementation

The implementation consists of four main components. The STM32 firmware communicates with the BH1750 sensor through the I²C interface and periodically acquires illumination measurements.

Group 4: Light Intensity Monitoring

A Python application receives the transmitted data through the USB Virtual COM Port and converts them into a structured JSON format. The Node.js backend server validates the received messages and distributes them to connected web clients through WebSocket communication. Finally, the frontend application updates the graphical user interface and visualizes the measurements in real time.

5.1 STM32 Firmware

The STM32 firmware is responsible for communicating with the BH1750 light sensor and acquiring illumination measurements periodically. The firmware is developed using the STM32 Hardware Abstraction Layer (HAL), which provides high-level functions for peripheral initialization and communication.

During system initialization, the I²C peripheral is configured and the BH1750 sensor is placed into Continuous High Resolution Mode. This operating mode enables the sensor to perform continuous measurements without requiring repeated configuration commands after each conversion.

Listing 1: Main measurement loop of the STM32 firmware.

```

1 BH1750_ReadData(); // Read sensor measurement
2
3 raw = ((uint16_t)buffer[0] << 8) | buffer[1];
4 // Combine MSB and LSB into a 16-bit value
5
6 lux = raw / 1.2f; // Convert raw value to lux
7
8 int length = sprintf(uartBuffer, "%.2f\r\n", lux);
9 // Format the lux value as an ASCII string
10
11 HAL_UART_Transmit(&huart2,
12                  (uint8_t *)uartBuffer,
13                  length,
14                  HAL_MAX_DELAY);
15 // Transmit the formatted string to the PC
16
17 HAL_Delay(1000); // Wait 1 second

```

Listing 1 presents the main execution loop of the STM32 firmware. First, the microcontroller reads the two-byte measurement from the BH1750 sensor through the I²C interface. The received most significant byte (MSB) and least significant byte (LSB) are combined into a single 16-bit unsigned integer representing the raw sensor output. The firmware then converts this raw value into illumination (lux) using the conversion relationship introduced in Equation (1).

The calculated illumination value is formatted as an ASCII string with two decimal places using the *sprintf()* function before being transmitted to the host computer via the UART interface. Finally, the firmware waits for one second before acquiring the next measurement, resulting in a sampling rate of approximately 1 Hz.

5.2 Python Serial Reader

The Python Serial Reader serves as the communication bridge between the STM32 microcontroller and the backend server. The application is implemented using the `pySerial` library for serial com-

munication and the `websocket-client` library for WebSocket communication.

The STM32 periodically transmits illumination measurements as ASCII strings terminated by a newline character. Upon receiving a serial packet, the Python application decodes the incoming bytes using UTF-8 encoding and converts the resulting string into a floating-point number. Invalid packets, decoding failures, or malformed numerical values are discarded to prevent corrupted data from propagating through the communication pipeline.

After validation, the illumination value is serialized using the `json.dumps()` function before being transmitted to the backend server through a persistent WebSocket connection.

Listing 2: Transmission of validated sensor data to the backend server.

```

1 lux = ser.readline()           # Read one UART packet
2 payload = into_json(lux)       # Validate and convert to float
3
4 if payload is not None:
5     json_output = json.dumps(payload) # Serialize into JSON format
6     ws.send(json_output)           # Send to the server

```

As shown in [Listing 2](#), the application first reads one measurement from the serial interface and converts it into a floating-point value using the `into_json()` function. This implementation keeps the Python application lightweight by delegating message formatting and client broadcasting to the backend server.

5.3 Backend Server

The Backend server serves as the communication bridge between the Python Serial Reader and the web-based graphical user interface. It is responsible for hosting the frontend application, receiving illumination measurements from the Python client through a WebSocket connection, validating the received data, and forwarding the measurements to all connected browser clients.

The backend is implemented using the Express framework to provide HTTP services and the `ws` library to implement WebSocket communication. The Express server hosts the static frontend resources, including the HTML, CSS, and JavaScript files, while the WebSocket server enables real-time bidirectional communication between the backend and connected clients.

After successful validation, the illumination value is serialized and immediately broadcast to every browser currently connected to the WebSocket server.

Listing 3: Validation and broadcasting of sensor measurements in the Node.js backend.

```

1 lux = JSON.parse(data.toString()); // Parse received message
2
3 if (!Number.isFinite(Number(lux)) ||
4     Number(lux) < 0) {
5     return; // Ignore invalid data
6 }
7
8 const sensorMessage =
9     JSON.stringify(Number(lux)); // Serialize measurement
10
11 for (const client of wss.clients) {

```


Group 4: Light Intensity Monitoring

```

12     if (client.readyState === WebSocket.OPEN) {
13         client.send(sensorMessage);    // Broadcast to all browsers
14     }
15 }

```

Listing 3 illustrates the core functionality of the Node.js backend. After receiving a message from the Python Serial Reader, the server parses the incoming data and verifies that it represents a valid non-negative numerical illumination value. The validated measurement is then serialized and broadcast to every browser currently connected to the WebSocket server. This approach enables multiple users to observe the same sensor measurements simultaneously while maintaining a simple and efficient communication architecture.

5.4 Web-Based Frontend

As described in [Section 4, User Interface](#), the frontend establishes a WebSocket connection with the backend server, receives incoming sensor measurements, and updates the displayed information dynamically without requiring the webpage to be refreshed.

Upon loading the webpage, the JavaScript application creates a persistent WebSocket connection to the backend server. This communication channel enables the frontend to receive new illumination measurements immediately after they are broadcast by the backend server.

Whenever a new sensor measurement is received, the frontend parses the incoming message and updates the displayed illumination value.

Listing 4: Processing incoming WebSocket messages in the frontend.

```

1 ws.addEventListener("message", (event) => {
2
3     let lux;
4
5     try {
6         lux = JSON.parse(event.data);    // Parse the received lux value
7     }
8     catch (err) {
9         console.error("Received invalid JSON:", event.data);
10        return;    // Ignore malformed packets
11    }
12
13    if (typeof lux === "number") {
14        appendPoint(new Date(), lux);    // Update the real-time graph
15    }
16    else {
17        console.warn("Unhandled message type");
18    }
19
20 });

```

Listing 4 illustrates the processing performed when a new WebSocket message is received by the frontend application. First, the received message is parsed into a numerical illumination value using the `JSON.parse()` function. If the received data are malformed, the packet is discarded to prevent invalid information from being displayed.

After successful parsing, the application verifies that the received value is of numerical type before updating the graphical visualization. The current local system time is recorded and associated with the received measurement, while the illumination value is appended to the Plotly [3] time-series graph.

Listing 5: Updating the real-time Plotly graph.

```

1 export function appendPoint(timestamp, value) {
2
3   // Use the received timestamp or generate the current time
4   const raw = timestamp !== undefined
5     ? timestamp
6     : new Date().toISOString();
7
8   // Convert the timestamp to UTC+7 format
9   const parts = getUTC7Parts(raw);
10  const x = partsToPlotlyISOString(parts);
11
12  // Store the new measurement locally
13  state.xData.push(x);
14  state.yData.push(value);
15
16  // Append the new point to the existing Plotly graph
17  Plotly.extendTraces(
18    plotDiv,
19    { x: [[x]], y: [[value]] },
20    [0]
21  );
22
23  // Refresh the numerical reading panel
24  updateReadingPanel(parts, value);
25 }

```

Listing 5 illustrates the procedure used to update the graphical user interface whenever a new illumination measurement is received. First, the timestamp is converted into the local time zone (UTC+7) before being appended to the horizontal axis of the graph. The corresponding illumination value is then stored together with the timestamp.

The Plotly function `extendTraces()` appends the new data point to the existing graph without redrawing the entire figure, thereby enabling efficient real-time visualization. Finally, the numerical display panel is refreshed to present the latest lux value and acquisition time to the user.

6 Results and Discussion

6.1 Experimental Setup and Procedure

The proposed monitoring system was evaluated under normal indoor lighting conditions. The experimental setup consisted of an STM32 microcontroller connected to a BH1750 digital light intensity sensor through the I²C interface. The STM32 periodically measured the ambient illumination and transmitted the acquired values to a host computer via a USB Virtual COM Port.

To evaluate the proposed monitoring system, the following procedure was carried out.

1. First, the Node.js backend server was started from a terminal using the command `node`

`server.js`. This launched both the HTTP server and the WebSocket server, making the web application available on the local host.

```
C:\Work and Study\Vinh_WS\Computer Network Project\Test02> node server.js
HTTP server: http://localhost:3000
|
```

Figure 4: Starting the Node.js backend server.

2. Next, the Python Serial Reader was executed using the command `python read_sensor.py`. The application established a serial connection with the STM32 through the USB Virtual COM Port, continuously received illumination measurements from the BH1750 sensor, and forwarded the validated measurements to the backend server through a WebSocket connection.

```
C:\Work and Study\Vinh_WS\Computer Network Project\Test02> python read_sensor.py
Connecting to COM3...
STM32 connected.
Connecting to WebSocket server at ws://localhost:3000/...
WebSocket connected.
```

Figure 5: Running the Python Serial Reader and establishing the WebSocket connection.

3. Finally, the local web server was exposed to external clients using **ngrok** [4], a globally distributed reverse proxy. The HTTP endpoint `http://localhost:3000` was tunneled through ngrok, generating a public URL that allowed remote users to access the web interface and receive real-time sensor measurements.

```
C:\Work and Study\Vinh_WS\Computer Network Project\Test02>ngrok http 3000|
```

```
ngrok (Ctrl+C to quit)
Request early access to new features: https://dashboard.ngrok.com/early-access

Session Status      online
Account             vinhbui2609@gmail.com (Plan: Free)
Version             3.39.8-msix-stable
Region              Asia Pacific (ap)
Web Interface       http://127.0.0.1:4040
Forwarding           https://murkiness-viewable-unlawful.ngrok-free.dev -> http://localhost:3000

Connections          ttl    opn    rt1    rt5    p50    p90
0                   0      0.00   0.00   0.00   0.00
```

Figure 6: Publishing the local web server using ngrok.

By exposing the local HTTP server through a secure public URL, remote users were able to access the graphical user interface from different devices while receiving the same real-time sensor measurements. This confirms that the proposed architecture can support Internet-based monitoring without requiring modifications to the application software.

6.2 Real-Time Monitoring Results

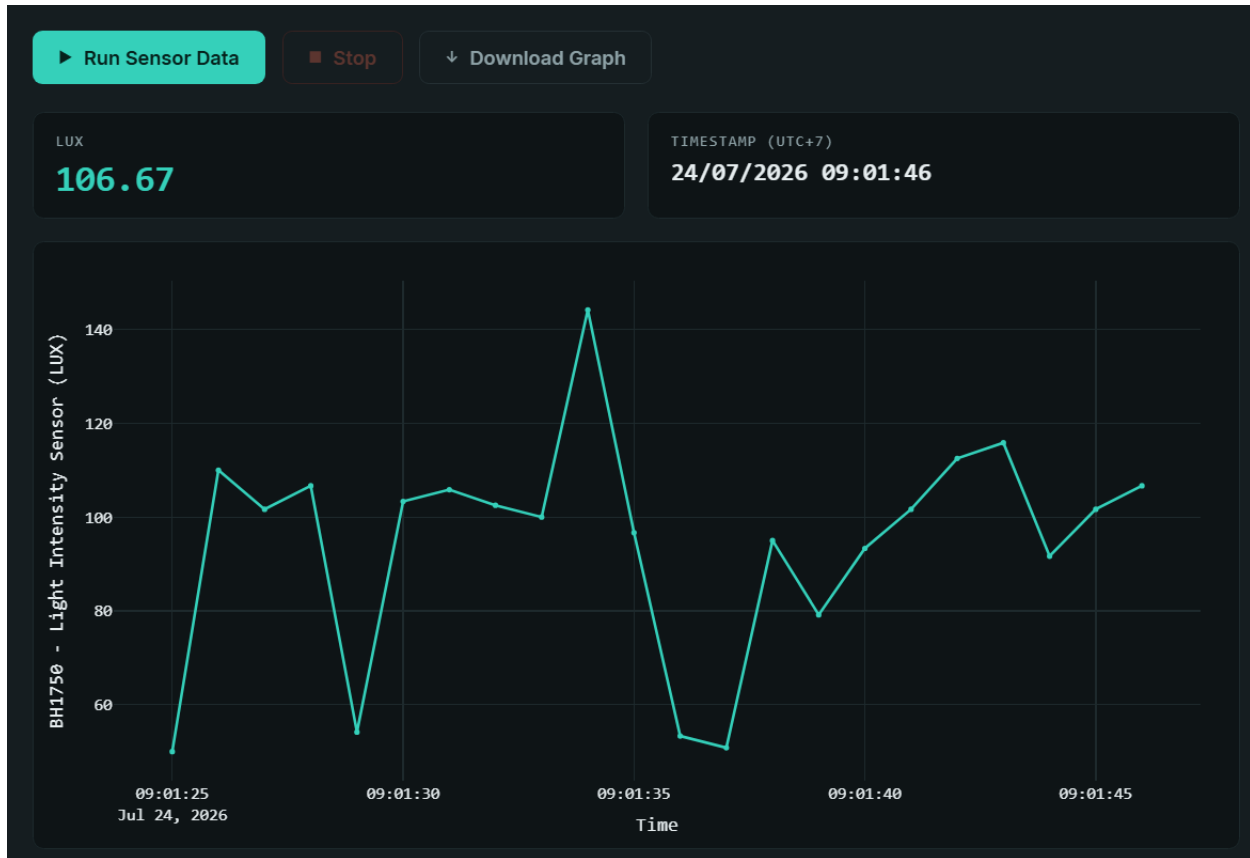
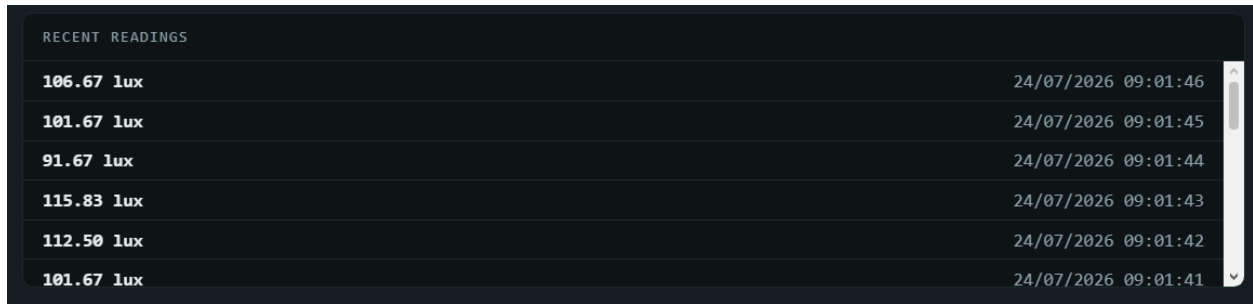


Figure 7: Main web-based graphical user interface during real-time monitoring.

Figure 7 illustrates the web interface during real-time monitoring. The upper information panel displays the latest illumination measurement acquired from the BH1750 sensor. The measured light intensity is presented in units of lux (lx) and is updated automatically whenever a new sensor reading is received from the backend server.

Alongside the measured lux value, the interface displays the timestamp corresponding to the latest received measurement. The timestamp is presented using the local time zone (UTC+7), allowing users to identify the exact time at which each measurement was recorded. Both the sensor value and timestamp are refreshed continuously, providing users with immediate feedback on the current lighting conditions.

The lower section of the interface presents a real-time line graph that visualizes the variation of light intensity over time. Each newly received measurement is appended to the graph immediately after arrival, enabling continuous monitoring of ambient illumination. This graphical representation allows users to observe both instantaneous changes and overall trends in the measured light intensity more effectively than numerical values alone.



RECENT READINGS	
106.67 lux	24/07/2026 09:01:46
101.67 lux	24/07/2026 09:01:45
91.67 lux	24/07/2026 09:01:44
115.83 lux	24/07/2026 09:01:43
112.50 lux	24/07/2026 09:01:42
101.67 lux	24/07/2026 09:01:41

Figure 8: Recent sensor readings displayed in chronological order.

Figure 8 shows the recent readings panel of the web interface. Each entry contains the measured illumination value in lux together with the timestamp indicating when the measurement was acquired.

The table is updated dynamically as new sensor data arrive, with the most recent measurement displayed at the top of the list. This feature allows users to review previously recorded values without relying solely on the graphical visualization. The combination of numerical records and graphical trends improves the usability of the monitoring system by providing both precise measurements and an overview of changes in ambient light intensity.

7 Conclusion

This project presented the design and implementation of a real-time light intensity monitoring system using the BH1750 digital light sensor, an STM32 microcontroller, and a web-based graphical user interface. The proposed system successfully integrates embedded hardware and modern web technologies to provide continuous monitoring of ambient illumination through a standard web browser.

The STM32 microcontroller periodically acquires illumination measurements from the BH1750 sensor via the I²C interface and transmits the measured values to a host computer through a USB Virtual COM Port. A Python application receives the serial data, validates the measurements, and forwards them to a Node.js backend server using a WebSocket connection. The backend subsequently distributes the received measurements to connected web clients, where the frontend dynamically updates the numerical display and the Plotly time-series graph in real time.

Experimental results demonstrated that the complete communication pipeline operated reliably, enabling low-latency transmission of sensor measurements from the hardware to the web interface. The modular software architecture also proved effective by separating the embedded firmware, serial communication, backend processing, and frontend visualization into independent components, thereby simplifying development, testing, and future maintenance.

8 ANNEX

8.1 read_sensor.py

```
1 import serial
2 import time
3 import json
4
5 import websocket
6
7 PORT = "COM3"
8 BAUD_RATE = 9600
9 TIMEOUT = 1
10 WS_URL = "ws://localhost:3000/"
11
12
13 def connect_serial():
14     print(f"Connecting to {PORT}...")
15
16     ser = serial.Serial(
17         port=PORT,
18         baudrate=BAUD_RATE,
19         parity=serial.PARITY_NONE,
20         stopbits=serial.STOPBITS_ONE,
21         bytesize=serial.EIGHTBITS,
22         timeout=TIMEOUT
23     )
24
25     print("STM32 connected.")
26     return ser
27
28
29 def into_json(raw_bytes):
30     try:
31         text = raw_bytes.decode("utf-8").strip()
32
33         if not text:
34             return None
35
36         value = float(text)
37         return value
38
39     except UnicodeDecodeError:
40         print("WARNING: UTF-8 decoding failed.")
41         return None
42
43     except ValueError:
44         print(f"WARNING: Invalid numeric value received: '{text}'")
45         return None
46
47
48
49 def open_websocket():
50     while True:
51         try:
52             print(f"Connecting to WebSocket server at {WS_URL}...")
53             ws = websocket.create_connection(WS_URL, timeout=5)
54             print("WebSocket connected.")
```

Group 4: Light Intensity Monitoring

```

55         return ws
56     except (OSError, websocket.WebSocketException) as error:
57         print(f"WebSocket connection failed: {error}")
58         time.sleep(2)
59
60
61 def main():
62     ser = None
63     ws = None
64
65     while True:
66         try:
67             if ser is None or not ser.is_open:
68                 ser = connect_serial()
69
70             if ws is None or not ws.connected:
71                 ws = open_websocket()
72
73             lux = ser.readline()
74             payload = into_json(lux)
75
76             if payload is not None:
77                 json_output = json.dumps(payload)
78                 ws.send(json_output)
79
80         except serial.SerialException as error:
81             print(f"Serial connection lost: {error}")
82             if ser is not None:
83                 ser.close()
84             ser = None
85             time.sleep(2)
86
87         except (websocket.WebSocketException, OSError) as error:
88             print(f"WebSocket connection lost: {error}")
89             if ws is not None:
90                 try:
91                     ws.close()
92                 except Exception:
93                     pass
94             ws = None
95             time.sleep(2)
96
97         except KeyboardInterrupt:
98             if ser is not None and ser.is_open:
99                 ser.close()
100             if ws is not None:
101                 ws.close()
102             print("\nProgram stopped.")
103             break
104
105
106 if __name__ == "__main__":
107     main()

```

8.2 server.js

```

1 const express = require("express");

```

```
2 const path = require("path");
3 const http = require("http");
4 const WebSocket = require("ws");
5
6 const app = express();
7 const PORT = 3000;
8 const ROOT = __dirname;
9
10 app.use(express.static(path.join(ROOT, "public")));
11
12 app.get("/", (req, res) => {
13   res.sendFile(path.join(ROOT, "public", "index.html"));
14 });
15
16 const server = http.createServer(app);
17 const wss = new WebSocket.Server({ server });
18
19 wss.on("connection", (socket, request) => {
20   const host = request.headers.host || 'localhost:${PORT}';
21   console.log("WebSocket connected");
22
23   socket.on("message", (data) => {
24     let lux;
25
26     try {
27       lux = JSON.parse(data.toString());
28     } catch {
29       return;
30     }
31
32     if (!Number.isFinite(Number(lux)) || Number(lux) < 0) {
33       return;
34     }
35
36     const sensorMessage = JSON.stringify(Number(lux));
37
38     console.log("Lux:", lux);
39
40     for (const client of wss.clients) {
41       if (client.readyState === WebSocket.OPEN) {
42         client.send(sensorMessage);
43       }
44     }
45   });
46
47   socket.on("close", () => {
48     console.log('WebSocket disconnected: ${socket.role}');
49   });
50
51   socket.on("error", (error) => {
52     console.error("WebSocket error:", error.message);
53   });
54 });
55
56 server.listen(PORT, () => {
57   console.log('HTTP server: http://localhost:${PORT}');
58 });
```


8.3 home.js

```

1 // --- Imports -----
2 import { runBtn, stopBtn, downloadBtn, downloadDropdown, downloadMenu } from "./
   home-DOM.js";
3 import { initPlot, downloadGraph } from "./home-plotly.js";
4 import { connectAndStart, stopStreaming } from "./home-WebSocket.js";
5
6 initPlot();
7
8 // --- Event wiring -----
9 runBtn.addEventListener("click", connectAndStart);
10 stopBtn.addEventListener("click", stopStreaming);
11
12 // --- Dropdown menu behavior -----
13 // Open menu
14 downloadBtn.addEventListener("click", (event) => {
15   event.stopPropagation();
16   downloadMenu.classList.toggle("is-open");
17 });
18
19 // Close menu
20 document.addEventListener("click", (event) => {
21   if (!downloadDropdown.contains(event.target)) {
22     downloadMenu.classList.remove("is-open");
23   }
24 });
25
26 // Format Selected -> get format type from HTML
27 downloadMenu.querySelectorAll(".dropdown__item").forEach((item) => {
28   item.addEventListener("click", () => {
29     downloadMenu.classList.remove("is-open");
30     downloadGraph(item.dataset.format);
31   });
32 });

```

8.4 home-WebSocket.js

```

1 // --- Imports -----
2 import {
3   runBtn,
4   stopBtn,
5   downloadBtn,
6   statusEl,
7   statusDot,
8   statusLabel,
9   hint,
10  plotDiv,
11  currentLuxEl,
12  currentTimeEl,
13  logListEl
14 } from "./home-DOM.js"
15
16 import {state, initPlot, appendPoint} from "./home-plotly.js"
17
18 // --- Configuration -----
19 // Matches server.js: http.createServer + WebSocket.Server on the same port.

```

```

20 const WS_URL = "ws://localhost:3000";
21
22 // --- UI state helpers -----
23 function setStatus(mode, label) {
24   statusEl.classList.remove("is-live", "is-error");
25   if (mode === "live") statusEl.classList.add("is-live");
26   if (mode === "error") statusEl.classList.add("is-error");
27   statusLabel.textContent = label;
28 }
29
30 function setStreamingUI(isStreaming) {
31   state.streaming = isStreaming;
32   runBtn.disabled = isStreaming;
33   stopBtn.disabled = !isStreaming;
34   hint.innerHTML = isStreaming
35     ? "Streaming live sensor data&hellip; click <strong>Stop</strong> to end the
       session."
36     : "Click <strong>Run Sensor Data</strong> to start streaming.";
37 }
38
39 // --- WebSocket handling -----
40 export function connectAndStart() {
41   setStatus("connecting", "Connecting");
42   runBtn.disabled = true; // prevent double-clicks while connecting
43
44   const ws = new WebSocket(WS_URL);
45   state.ws = ws;
46
47   ws.addEventListener("open", () => {
48     setStatus("live", "Streaming");
49     setStreamingUI(true);
50     // Reset the graph for a fresh run
51     state.xData = [];
52     state.yData = [];
53     initPlot();
54     currentLuxEl.textContent = "\u2014";
55     currentTimeEl.textContent = "\u2014";
56     logListEl.innerHTML = "";
57   });
58
59   ws.addEventListener("message", (event) => {
60     let lux;
61
62     try {
63       lux = JSON.parse(event.data); // event.data is a JSON number, e.g. 123.45
64     } catch (err) {
65       console.error("Received invalid JSON:", event.data);
66       return;
67     }
68
69     if (typeof lux === "number") {
70       appendPoint(new Date(), lux);
71     } else {
72       console.warn("Unhandled message type:");
73     }
74   });
75
76   ws.addEventListener("close", () => {
77     setStatus("idle", "Disconnected");

```

```

78     setStreamingUI(false);
79     downloadBtn.disabled = state.xData.length === 0;
80   });
81
82   ws.addEventListener("error", (err) => {
83     console.error("WebSocket error:", err);
84     setStatus("error", "Connection error");
85   });
86 }
87
88 export function stopStreaming() {
89   if (state.ws && state.ws.readyState === WebSocket.OPEN) {
90     state.ws.close();
91   }
92   setStreamingUI(false);
93   setStatus("idle", "Disconnected");
94   downloadBtn.disabled = state.xData.length === 0;
95 }

```

8.5 home-DOM.js

```

1  // --- DOM references -----
2
3  export const runBtn = document.getElementById("runBtn");
4
5  export const stopBtn = document.getElementById("stopBtn");
6
7  export const downloadBtn = document.getElementById("downloadBtn");
8
9  export const downloadMenu = document.getElementById("downloadMenu");
10
11 export const downloadDropdown = document.getElementById("downloadDropdown");
12
13 export const statusEl = document.getElementById("status");
14
15 export const statusDot = document.getElementById("statusDot");
16
17 export const statusLabel = document.getElementById("statusLabel");
18
19 export const hint = document.getElementById("hint");
20
21 export const plotDiv = document.getElementById("plot");
22
23 export const currentLuxEl = document.getElementById("currentLux");
24
25 export const currentTimeEl = document.getElementById("currentTime");
26
27 export const logListEl = document.getElementById("logList");

```

8.6 home-plotly.js

```

1  // --- Imports -----
2  import { getUTC7Parts, partsToPlotlyISOString, formatUTC7Display } from "../home-
3  |   time_zone.js";
4  import {

```

```

4   plotDiv,
5   currentLuxEl,
6   currentTimeEl,
7   logListEl
8 } from "../home-DOM.js"
9
10  // --- Plotly setup -----
11  const plotLayout = {
12    paper_bgcolor: "transparent",
13    plot_bgcolor: "transparent",
14    font: { color: "#e7eef0", family: "IBM Plex Mono, monospace", size: 12 },
15    margin: { l: 50, r: 20, t: 20, b: 40 },
16    xaxis: { title: "Time", gridcolor: "#1e2a2d", zeroline: false },
17    yaxis: { title: "BH1750 - Light Intensity Sensor (LUX)", gridcolor: "#1e2a2d",
18              zeroline: false },
19    showlegend: false,
20  };
21
22  const plotConfig = { responsive: true, displayModeBar: false };
23
24  // --- Local state (per tab/session) -----
25  export const state = {
26    ws: null,
27    streaming: false,
28    xData: [],
29    yData: [],
30  };
31
32  export function initPlot() {
33    const trace = {
34      x: [],
35      y: [],
36      mode: "lines+markers",
37      type: "scatter",
38      line: { color: "#35d0ba", width: 2 },
39      marker: { size: 4, color: "#35d0ba" },
40    };
41    Plotly.newPlot(plotDiv, [trace], plotLayout, plotConfig);
42  }
43
44  // --- Graph and Panel updates -----
45
46  export function appendPoint(timestamp, value) {
47    const raw = timestamp !== undefined ? timestamp : new Date().toISOString();
48    const parts = getUTC7Parts(raw);
49    const x = partsToPlotlyISOString(parts);
50
51    state.xData.push(x);
52    state.yData.push(value);
53
54    Plotly.extendTraces(plotDiv, { x: [[x]], y: [[value]] }, [0]);
55
56    // Keep only the most recent N points visible to avoid the graph
57    // becoming unreadable during long streams. Adjust as needed.
58    const MAX_VISIBLE = 200;
59    if (state.xData.length > MAX_VISIBLE) {
60      const excess = state.xData.length - MAX_VISIBLE;
61      Plotly.relayout(plotDiv, {
62        "xaxis.range": [state.xData[excess], x],

```

```

61     });
62   }
63
64   updateReadingPanel(parts, value);
65 }
66
67 function updateReadingPanel(parts, value) {
68   const formattedTime = formatUTC7Display(parts);
69
70   currentLuxEl.textContent = value.toFixed(2);
71   currentTimeEl.textContent = formattedTime;
72
73   const row = document.createElement("div");
74   row.className = "log__row";
75   row.innerHTML = `<span class="log__lux">${value.toFixed(2)} lux</span><span>${
76     formattedTime}</span>`;
77   logListEl.prepend(row);
78
79   // Cap the log so it doesn't grow unbounded during a long stream.
80   const MAX_LOG_ROWS = 50;
81   while (logListEl.children.length > MAX_LOG_ROWS) {
82     logListEl.removeChild(logListEl.lastChild);
83   }
84 }
85
86 // --- Graph export (SVG / PNG / JSON) -----
87
88 function buildExportLayout() {
89   return {
90     ...plotLayout,
91     paper_bgcolor: "#ffffff",
92     plot_bgcolor: "#ffffff",
93     font: { ...plotLayout.font, color: "#111111" },
94     axis: {
95       ...plotLayout.axis,
96       gridcolor: "#cccccc",
97       zerolinecolor: "#cccccc",
98       linecolor: "#111111",
99       tickfont: { color: "#111111" },
100     },
101     yaxis: {
102       ...plotLayout.yaxis,
103       gridcolor: "#cccccc",
104       zerolinecolor: "#cccccc",
105       linecolor: "#111111",
106       tickfont: { color: "#111111" },
107     },
108   };
109 }
110
111 async function downloadImageFormat(format) {
112   const hiddenDiv = document.createElement("div");
113   hiddenDiv.style.position = "fixed";
114   hiddenDiv.style.top = "0";
115   hiddenDiv.style.left = "-9999px";
116   hiddenDiv.style.width = "900px";
117   hiddenDiv.style.height = "500px";
118   document.body.appendChild(hiddenDiv);

```

```

119     try {
120         await Plotly.newPlot(hiddenDiv, plotDiv.data, buildExportLayout(), {
121             staticPlot: true,
122         });
123         await Plotly.downloadImage(hiddenDiv, {
124             format,
125             filename: "sensor-graph",
126             width: 900,
127             height: 500,
128         });
129     } finally {
130         Plotly.purge(hiddenDiv);
131         hiddenDiv.remove();
132     }
133 }
134
135 function downloadJSON() {
136     const exportObject = {
137         data: plotDiv.data,
138         layout: plotDiv.layout,
139     };
140     const blob = new Blob([JSON.stringify(exportObject, null, 2)], {
141         type: "application/json",
142     });
143     triggerDownload(blob, "sensor-graph.json");
144 }
145
146 function triggerDownload(blob, filename) {
147     const url = URL.createObjectURL(blob);
148     const a = document.createElement("a");
149     a.href = url;
150     a.download = filename;
151     document.body.appendChild(a);
152     a.click();
153     a.remove();
154     URL.revokeObjectURL(url);
155 }
156
157 export function downloadGraph(format) {
158     if (format === "json") {
159         downloadJSON();
160     } else {
161         downloadImageFormat(format);
162     }
163 }

```

8.7 home-time_zone.js

```

1  // --- Timezone handling (UTC+7) -----
2  const UTC7_PART_FORMATTER = new Intl.DateTimeFormat("en-US", {
3      timeZone: "Asia/Ho_Chi_Minh",
4      year: "numeric",
5      month: "2-digit",
6      day: "2-digit",
7      hour: "2-digit",
8      minute: "2-digit",
9      second: "2-digit",

```

```

10   hour12: false,
11 });
12
13 export function getUTC7Parts(timestamp) {
14   const date = new Date(timestamp);
15   const parts = UTC7_PART_FORMATTER.formatToParts(date).reduce((acc, p) => {
16     acc[p.type] = p.value;
17     return acc;
18   }, {});
19   if (parts.hour === "24") parts.hour = "00";
20   return parts;
21 }
22
23 export function partsToPlotlyISOString(parts) {
24   return `${parts.year}-${parts.month}-${parts.day}T${parts.hour}:${parts.minute}:
25     ${parts.second}.000Z`;
26 }
27
28 export function formatUTC7Display(parts) {
29   return `${parts.day}/${parts.month}/${parts.year} ${parts.hour}:${parts.minute}:
30     ${parts.second}`;
31 }

```

8.8 index.html

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>Sensor Data Stream</title>
7   <link rel="stylesheet" href="style.css" />
8 </head>
9 <body>
10   <div class="app">
11     <header class="app__header">
12       <div class="app__title">
13         <span class="app__eyebrow">Live Telemetry</span>
14         <h1>Sensor Data Stream</h1>
15       </div>
16       <div class="status" id="status">
17         <span class="status__dot" id="statusDot"></span>
18         <span class="status__label" id="statusLabel">Disconnected</span>
19       </div>
20     </header>
21
22     <main class="app__main">
23       <section class="panel">
24         <div class="panel__controls">
25           <button id="runBtn" class="btn btn--primary">
26             <span class="btn__icon">&#9654;</span> Run Sensor Data
27           </button>
28           <button id="stopBtn" class="btn btn--danger disabled">
29             <span class="btn__icon">&#9632;</span> Stop
30           </button>
31           <div class="dropdown" id="downloadDropdown">
32             <button id="downloadBtn" class="btn btn--ghost disabled">

```

```

33     <span class="btn__icon">&#8595;</span> Download Graph
34   </button>
35   <div class="dropdown__menu" id="downloadMenu">
36     <button class="dropdown__item" data-format="png">PNG image</button>
37     <button class="dropdown__item" data-format="svg">SVG</button>
38     <button class="dropdown__item" data-format="json">JSON (raw data)</
39       button>
40   </div>
41 </div>
42
43 <div class="reading">
44   <div class="reading__block">
45     <span class="reading__label">Lux</span>
46     <span class="reading__value" id="currentLux">&mdash;</span>
47   </div>
48   <div class="reading__block">
49     <span class="reading__label">Timestamp (UTC+7)</span>
50     <span class="reading__value reading__value--time" id="currentTime">&
51       mdash;</span>
52   </div>
53 </div>
54
55 <div class="panel__graph">
56   <div id="plot"></div>
57 </div>
58
59 <div class="log">
60   <div class="log__header">Recent readings</div>
61   <div class="log__list" id="logList"></div>
62 </div>
63
64 <p class="panel__hint" id="hint">
65   Click <strong>Run Sensor Data</strong> to start streaming.
66 </p>
67 </section>
68 </main>
69 </div>
70
71 <script src="https://cdn.plot.ly/plotly-2.32.0.min.js"></script>
72 <script type = "module" src="home.js"></script>
73 </body>
74 </html>

```

8.9 style.css

```

1 :root {
2   --bg: #0e1416;
3   --panel: #151d20;
4   --panel-border: #253034;
5   --text: #e7eef0;
6   --text-dim: #8fa3a8;
7   --accent: #35d0ba;
8   --accent-dim: #1f8f7e;
9   --danger: #e0654f;
10  --danger-dim: #7a3229;
11  --grid: #1e2a2d;

```


Group 4: Light Intensity Monitoring

```

12  --mono: "IBM Plex Mono", "SFMono-Regular", Consolas, monospace;
13  --sans: "Inter", -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
14  }
15
16  * { box-sizing: border-box; }
17
18  html, body {
19    margin: 0;
20    padding: 0;
21    background: var(--bg);
22    color: var(--text);
23    font-family: var(--sans);
24    min-height: 100vh;
25  }
26
27  .app {
28    max-width: 960px;
29    margin: 0 auto;
30    padding: 32px 20px 60px;
31  }
32
33  .app__header {
34    display: flex;
35    justify-content: space-between;
36    align-items: flex-end;
37    border-bottom: 1px solid var(--panel-border);
38    padding-bottom: 20px;
39    margin-bottom: 24px;
40    flex-wrap: wrap;
41    gap: 12px;
42  }
43
44  .app__eyebrow {
45    display: block;
46    font-family: var(--mono);
47    font-size: 12px;
48    letter-spacing: 0.12em;
49    text-transform: uppercase;
50    color: var(--accent);
51    margin-bottom: 6px;
52  }
53
54  .app__title h1 {
55    margin: 0;
56    font-size: 28px;
57    font-weight: 600;
58    letter-spacing: -0.01em;
59  }
60
61  .status {
62    display: flex;
63    align-items: center;
64    gap: 8px;
65    font-family: var(--mono);
66    font-size: 13px;
67    color: var(--text-dim);
68  }
69
70  .status__dot {

```

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```

71   width: 9px;
72   height: 9px;
73   border-radius: 50%;
74   background: var(--text-dim);
75   transition: background 0.2s ease, box-shadow 0.2s ease;
76 }
77
78 .status.is-live .status__dot {
79   background: var(--accent);
80   box-shadow: 0 0 8px var(--accent);
81   animation: pulse 1.4s ease-in-out infinite;
82 }
83
84 .status.is-error .status__dot {
85   background: var(--danger);
86   box-shadow: 0 0 8px var(--danger);
87 }
88
89 @keyframes pulse {
90   0%, 100% { opacity: 1; }
91   50% { opacity: 0.4; }
92 }
93
94 .panel {
95   background: var(--panel);
96   border: 1px solid var(--panel-border);
97   border-radius: 10px;
98   padding: 24px;
99 }
100
101 .panel__controls {
102   display: flex;
103   gap: 12px;
104   flex-wrap: wrap;
105   margin-bottom: 20px;
106 }
107
108 .dropdown {
109   position: relative;
110 }
111
112 .dropdown__menu {
113   display: none;
114   position: absolute;
115   top: calc(100% + 6px);
116   left: 0;
117   min-width: 180px;
118   background: var(--panel);
119   border: 1px solid var(--panel-border);
120   border-radius: 8px;
121   padding: 6px;
122   box-shadow: 0 8px 24px rgba(0, 0, 0, 0.4);
123   z-index: 10;
124 }
125
126 .dropdown__menu.is-open {
127   display: flex;
128   flex-direction: column;
129   gap: 2px;

```

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```

130 }
131
132 .dropdown__item {
133   background: transparent;
134   border: none;
135   color: var(--text);
136   font-family: var(--sans);
137   font-size: 13.5px;
138   text-align: left;
139   padding: 8px 10px;
140   border-radius: 5px;
141   cursor: pointer;
142 }
143
144 .dropdown__item:hover {
145   background: rgba(255, 255, 255, 0.06);
146   color: var(--accent);
147 }
148
149 .btn {
150   display: inline-flex;
151   align-items: center;
152   gap: 8px;
153   padding: 10px 18px;
154   font-family: var(--sans);
155   font-size: 14px;
156   font-weight: 600;
157   border-radius: 7px;
158   border: 1px solid transparent;
159   cursor: pointer;
160   transition: transform 0.08s ease, opacity 0.15s ease, background 0.15s ease;
161 }
162
163 .btn:active:not(:disabled) { transform: translateY(1px); }
164
165 .btn:disabled {
166   opacity: 0.35;
167   cursor: not-allowed;
168 }
169
170 .btn__icon { font-size: 11px; }
171
172 .btn--primary {
173   background: var(--accent);
174   color: #06201b;
175 }
176 .btn--primary:hover:not(:disabled) { background: #4ee0cb; }
177
178 .btn--danger {
179   background: transparent;
180   color: var(--danger);
181   border-color: var(--danger-dim);
182 }
183 .btn--danger:hover:not(:disabled) {
184   background: rgba(224, 101, 79, 0.1);
185 }
186
187 .btn--ghost {
188   background: transparent;

```

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```

189   color: var(--text-dim);
190   border-color: var(--panel-border);
191 }
192 .btn--ghost:hover:not(:disabled) {
193   color: var(--text);
194   border-color: var(--text-dim);
195 }
196
197 .panel__graph {
198   background: var(--bg);
199   border: 1px solid var(--grid);
200   border-radius: 8px;
201   padding: 8px;
202 }
203
204 .reading {
205   display: flex;
206   gap: 16px;
207   margin-bottom: 16px;
208   flex-wrap: wrap;
209 }
210
211 .reading__block {
212   flex: 1;
213   min-width: 180px;
214   background: var(--bg);
215   border: 1px solid var(--grid);
216   border-radius: 8px;
217   padding: 14px 16px;
218 }
219
220 .reading__label {
221   display: block;
222   font-family: var(--mono);
223   font-size: 11px;
224   letter-spacing: 0.08em;
225   text-transform: uppercase;
226   color: var(--text-dim);
227   margin-bottom: 6px;
228 }
229
230 .reading__value {
231   display: block;
232   font-family: var(--mono);
233   font-size: 26px;
234   font-weight: 600;
235   color: var(--accent);
236   line-height: 1.1;
237 }
238
239 .reading__value--time {
240   font-size: 18px;
241   color: var(--text);
242 }
243
244 .log {
245   margin-top: 16px;
246   background: var(--bg);
247   border: 1px solid var(--grid);

```

Group 4: Light Intensity Monitoring

```

248     border-radius: 8px;
249     overflow: hidden;
250 }
251
252 .log__header {
253     font-family: var(--mono);
254     font-size: 11px;
255     letter-spacing: 0.08em;
256     text-transform: uppercase;
257     color: var(--text-dim);
258     padding: 10px 14px;
259     border-bottom: 1px solid var(--grid);
260 }
261
262 .log__list {
263     max-height: 160px;
264     overflow-y: auto;
265     font-family: var(--mono);
266     font-size: 13px;
267 }
268
269 .log__row {
270     display: flex;
271     justify-content: space-between;
272     padding: 6px 14px;
273     border-bottom: 1px solid var(--grid);
274     color: var(--text-dim);
275 }
276
277 .log__row:last-child {
278     border-bottom: none;
279 }
280
281 .log__row .log__lux {
282     color: var(--text);
283     font-weight: 600;
284 }
285
286 #plot {
287     width: 100%;
288     height: 420px;
289 }
290
291 .panel__hint {
292     margin: 16px 2px 0;
293     font-family: var(--mono);
294     font-size: 12.5px;
295     color: var(--text-dim);
296 }
297
298 .panel__hint strong {
299     color: var(--text);
300     font-weight: 600;
301 }
302
303 @media (max-width: 600px) {
304     .app__header { flex-direction: column; align-items: flex-start; }
305     .panel__controls { flex-direction: column; }
306     .btn { width: 100%; justify-content: center; }

```

```

307 .dropdown { width: 100%; }
308 .dropdown__menu { left: 0; right: 0; min-width: unset; }
309 #plot { height: 320px; }
310 }

```

8.10 BH1750.h

```

1  #ifndef BH1750_H_
2  #define BH1750_H_
3
4  #include <stdint.h>
5
6  /* Macro Command */
7  // Slave address
8  #define ADDR_HIGH_ADDRESS_WRITE    (0xB8 + 0)
9  #define ADDR_HIGH_ADDRESS_READ     (0xB8 + 1)
10 #define ADDR_LOW_ADDRESS_WRITE     (0x46 + 0)
11 #define ADDR_LOW_ADDRESS_READ      (0x46 + 1)
12
13
14 // Commands table
15 #define BH1750_POWER_DOWN          0x00
16 #define BH1750_POWER_ON            0x01
17 #define BH1750_RESET                0x07
18 #define BH1750_CON_H_RES_MODE      0x10
19 #define BH1750_CON_H_RES_MODE2     0x11
20 #define BH1750_CON_L_RES_MODE      0x13
21 #define BH1750_ONE_H_RES_MODE      0x20
22 #define BH1750_ONE_H_RES_MODE2     0x21
23 #define BH1750_ONE_L_RES_MODE      0x23
24 #define BH1750_CHANGE_HIGH_MT      0x08    //01000_XXX
25 #define BH1750_CHANGE_LOW_MT       0x03    //011_XXXXX
26
27 // Sending data
28 #define POW_2_OF_15      32768
29 #define POW_2_OF_9       512
30 #define POW_2_OF_8       256
31 #define POW_2_OF_7       128
32 #define POW_2_OF_4       16
33
34
35 /* Function declaration */
36 void BH1750_ContinuousMode();
37 void BH1750_SendCommand(uint8_t cmd);
38 void BH1750_ReadData();
39
40
41 #endif

```

8.11 BH1750.c

```

1  #include "BH1750.h"
2  #include "i2c.h"
3
4  extern uint8_t buffer[2];

```

Group 4: Light Intensity Monitoring

```

5
6 // MODE pin -> LOW = Continuous Mode
7 void BH1750_ContinuousMode()
8 {
9     HAL_GPIO_WritePin(BH1750_MODE_GPIO_Port, BH1750_MODE_Pin, GPIO_PIN_RESET);
10 }
11
12
13 void BH1750_SendCommand(uint8_t cmd)
14 {
15     HAL_I2C_Master_Transmit(&hi2c1, ADDR_LOW_ADDRESS_WRITE, &cmd, 1, HAL_MAX_DELAY);
16 }
17
18
19 void BH1750_ReadData()
20 {
21     HAL_I2C_Master_Receive(&hi2c1, ADDR_LOW_ADDRESS_READ, buffer, 2, HAL_MAX_DELAY);
22 }

```

8.12 main.c

```

1 /* USER CODE BEGIN Header */
2 /**
3  * *****
4  * @file           : main.c
5  * @brief          : Main program body
6  * *****
7  * @attention
8  *
9  * Copyright (c) 2024 STMicroelectronics.
10 * All rights reserved.
11 *
12 * This software is licensed under terms that can be found in the LICENSE file
13 * in the root directory of this software component.
14 * If no LICENSE file comes with this software, it is provided AS-IS.
15 *
16 * *****
17 */
18 /* USER CODE END Header */
19 /* Includes -----*/
20 #include "main.h"
21 #include "i2c.h"
22 #include "usart.h"
23 #include "gpio.h"
24
25 /* Private includes -----*/
26 /* USER CODE BEGIN Includes */
27 #include "BH1750.h"
28 #include "usart.h"
29 #include <stdio.h>
30 /* USER CODE END Includes */
31
32 /* Private typedef -----*/
33 /* USER CODE BEGIN PTD */
34

```

Group 4: Light Intensity Monitoring

```

35  /* USER CODE END PTD */
36
37  /* Private define -----*/
38  /* USER CODE BEGIN PD */
39
40  /* USER CODE END PD */
41
42  /* Private macro -----*/
43  /* USER CODE BEGIN PM */
44
45  /* USER CODE END PM */
46
47  /* Private variables -----*/
48
49  /* USER CODE BEGIN PV */
50
51  // Buffer used to receive the two-byte light measurement from the BH1750
52  uint8_t buffer[2] = {0};
53
54  // Raw 16-bit sensor output
55  uint16_t raw = 0;
56
57  float lux = 0.0f;
58  char uartBuffer[64];
59
60  /* USER CODE END PV */
61
62  /* Private function prototypes -----*/
63  void SystemClock_Config(void);
64  /* USER CODE BEGIN PFP */
65
66  /* USER CODE END PFP */
67
68  /* Private user code -----*/
69  /* USER CODE BEGIN 0 */
70
71  /* USER CODE END 0 */
72
73  /**
74   * @brief The application entry point.
75   * @retval int
76   */
77  int main(void)
78  {
79
80      /* USER CODE BEGIN 1 */
81
82      /* USER CODE END 1 */
83
84      /* MCU Configuration-----*/
85
86      /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
87      HAL_Init();
88
89      /* USER CODE BEGIN Init */
90
91      /* USER CODE END Init */
92
93      /* Configure the system clock */

```



```

94  SystemClock_Config();
95
96  /* USER CODE BEGIN SysInit */
97
98  /* USER CODE END SysInit */
99
100 /* Initialize all configured peripherals */
101 MX_GPIO_Init();
102 MX_I2C1_Init();
103 MX_USART2_UART_Init();
104 /* USER CODE BEGIN 2 */
105
106 // Setup Reading Mode as Continuous
107 BH1750_ContinuousMode();
108
109 // Power on the sensor
110 BH1750_SendCommand(BH1750_POWER_ON);
111
112 // Clear previous measurement data
113 BH1750_SendCommand(BH1750_RESET);
114
115 // Start continuous High Resolution measurement mode
116 BH1750_SendCommand(BH1750_CON_H_RES_MODE);
117
118
119 /* USER CODE END 2 */
120
121 /* Infinite loop */
122 /* USER CODE BEGIN WHILE */
123 while (1)
124 {
125     // Read two bytes from BH1750
126     BH1750_ReadData();
127
128     /*-----
129     * Combine MSB and LSB into one 16-bit value
130     *
131     * buffer[0] = High byte
132     * buffer[1] = Low byte
133     *-----*/
134     raw = ((uint16_t)buffer[0] << 8) | buffer[1];
135
136     /*-----
137     * Convert raw measurement into Lux
138     *
139     * According to BH1750 datasheet:
140     *     Lux = Raw / 1.2
141     *-----*/
142     lux = raw / 1.2f;
143
144     int length = sprintf(uartBuffer, "%.2f\r\n", lux);
145
146     HAL_UART_Transmit(&huart2, (uint8_t *)uartBuffer, length, HAL_MAX_DELAY);
147
148     /* Wait before requesting another measurement */
149     HAL_Delay(1000);
150
151     /* USER CODE END WHILE */
152

```

Group 4: Light Intensity Monitoring

```

153     /* USER CODE BEGIN 3 */
154 }
155 /* USER CODE END 3 */
156 }
157
158 /**
159  * @brief System Clock Configuration
160  * @retval None
161  */
162 void SystemClock_Config(void)
163 {
164     RCC_OscInitTypeDef RCC_OscInitStruct = {0};
165     RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
166
167     /** Configure the main internal regulator output voltage
168     */
169     __HAL_RCC_PWR_CLK_ENABLE();
170     __HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE1);
171
172     /** Initializes the RCC Oscillators according to the specified parameters
173     * in the RCC_OscInitTypeDef structure.
174     */
175     RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
176     RCC_OscInitStruct.HSISState = RCC_HSI_ON;
177     RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
178     RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;
179     if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
180     {
181         Error_Handler();
182     }
183
184     /** Initializes the CPU, AHB and APB buses clocks
185     */
186     RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
187         |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
188     RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
189     RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
190     RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
191     RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;
192
193     if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_0) != HAL_OK)
194     {
195         Error_Handler();
196     }
197 }
198
199 /* USER CODE BEGIN 4 */
200
201 /* USER CODE END 4 */
202
203 /**
204  * @brief This function is executed in case of error occurrence.
205  * @retval None
206  */
207 void Error_Handler(void)
208 {
209     /* USER CODE BEGIN Error_Handler_Debug */
210     /* User can add his own implementation to report the HAL error return state */
211     __disable_irq();

```

```
212     while (1)
213     {
214     }
215     /* USER CODE END Error_Handler_Debug */
216 }
217 #ifndef USE_FULL_ASSERT
218 /**
219  * @brief Reports the name of the source file and the source line number
220  *        where the assert_param error has occurred.
221  * @param file: pointer to the source file name
222  * @param line: assert_param error line source number
223  * @retval None
224  */
225 void assert_failed(uint8_t *file, uint32_t line)
226 {
227     /* USER CODE BEGIN 6 */
228     /* User can add his own implementation to report the file name and line number,
229        ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
230     /* USER CODE END 6 */
231 }
232 #endif /* USE_FULL_ASSERT */
```

References

- [1] *Digital 16-bit Serial Output Type Ambient Light Sensor BH1750FVI*, ROHM Semiconductor, 2011. [Online]. Available: <https://www.mouser.com/datasheet/2/348/bh1750fvi-e-186247.pdf>
- [2] STMicroelectronics, *RM0383: STM32F411 Reference Manual*, 2023. [Online]. Available: https://www.st.com/resource/en/reference_manual/dm00119316.pdf
- [3] Plotly Technologies Inc., “Plotly javascript graphing library documentation,” <https://plotly.com/javascript/>.
- [4] ngrok Inc., “Get started with ngrok,” <https://ngrok.com/docs>.